

MTN GHANA TEKYEREMA PA HACKATHON 2026

Tie Me Ghana

A deaf patient and a doctor, talking to each other.
Today, on the phones they already carry.

Team [your team name]

Pilot. Not yet approved for clinical use.

Three doors, all closed

1

She cannot hear him

Speech is the hospital's default, and it carries nothing to a Deaf patient.

2

She cannot read it

Writing is the fallback everywhere. Many Deaf Ghanaians did not finish school, so the note is no better than the speech.

3

There is no interpreter

GhSL interpreters are few. A ward cannot staff one, and at night there is nobody at all.

So the visit runs on hand waving and guessing. The patient cannot say where the pain is. The doctor cannot be sure she understood the medicine. Both of them leave hoping.

WHAT IT IS

One web page, two ways to use it

No app store, no download, no new hardware on the ward.

Two devices

WORKING NOW

The patient holds her own phone. A six character code pairs them, and from then on the two devices talk straight to each other.

Or one device

WORKING NOW

No smart phone? The same visit runs on one screen, split down the middle, turned between the two people.

108

reviewed GhSL clips

13.2 MB

whole library, on the phone

0

consultation data on our server

It opens in any browser, on any phone, on whatever laptop the ward already owns. Add it to the home screen and it behaves like an app, with the sign videos stored on the device.

Working today

Every item below runs in the build you are about to see. Nothing here is a mock-up.

WORKING NOW

- Doctor speaks or types. The sentence is captioned in Twi and played as GhSL sign video.
- The literacy question is asked in sign, on the patient's own phone, before anything else.
- Guided Interrogation: the doctor taps a question, the patient answers Yes or No in sign.
- The patient types a reply and the doctor's device speaks it out loud.

WORKING NOW

- Emergency triage: four critical alerts, a body drawing and a pain scale, mirrored to her phone.
- Prescription as sign video plus a QR code, saved to the patient's phone and played offline.
- The safety gate refuses any sentence it cannot sign correctly, and names the missing word.
- Session transcript kept on each device. Pairing survives a reload of either phone.

1,933 automated tests. 61 written architecture decisions, including the mistakes. Deployed and reachable right now, not running on this laptop.

Planned, and honestly not built

We would rather list these than have you find them.

Fingerspelling

PLANNED

9 of 26 letters filmed. Spelling a word needs every one of its letters, so the fallback stays off and a word with no sign stops the sentence instead.

Body-part grid in Guided mode

PLANNED

0 of 16 filmed. The app hides the grid rather than offer three body parts when the pain is in a fourth — a wrong answer there looks exactly like a right one.

Clinical sign-off

PLANNED

The sign library and the word-safety lists need a GhSL consultant and a clinician. Until then it is a pilot, and every screen says so.

Direct connection on closed networks

PLANNED

No TURN server, so some hospital networks will refuse to pair two devices. The app says so and offers the one-screen mode instead of failing quietly.

Ga and Ewe

PLANNED

The language layer is an interface and clip lookup is keyed on gloss, not on Twi, so it extends — but it is not built.

Visual queue call

PLANNED

Designed against the vibration vocabulary already in the app. Not built.

Ten minutes, in this order

Screens 1 to 9. Times are budgets, not targets.

1 Does she have a phone? 40s	2 Pair with a six letter code 50s	3 Can she read? Asked in sign 50s
4 Guided questions, answered in sign 2m	5 Free typing, and a voice for the doctor 70s	6 The app refuses a sentence 60s
7 Emergency, mirrored to her phone 60s	8 The medicine, saved on her phone 2m	9 One shared screen, and no app store 80s

Running late? Cut 5 and 9. Never cut 6 or 8 — the refusal and the medicine are the two that win marks.

It would rather show nothing than half a sentence

“Your pain is not serious.”

Refused. The word “not” has no reviewed sign, so the sentence does not play.

A normal app drops the word it does not know and plays the rest. The patient would have read “your pain is serious” — the opposite of what the doctor said.

In a hospital that is how somebody takes the wrong dose. So every word is classified before anything plays: droppable, or safe to spell, or blocking. One blocking word stops the whole sentence and the doctor is told which word it was.

How a word is treated

- Has a reviewed sign → play it
- An article or copula → drop it, GhSL does not use them
- Blocking, like “not” → stop the sentence
- Otherwise → spell it, once the alphabet exists

Communication support and two-way interaction

Both directions, and neither of them needs an interpreter.

Doctor → patient

WORKING NOW

- Speech or typing, in English or Twi
- Caption shown in Twi, and played as GhSL sign video
- 14 whole questions filmed as single clips, so a common question is one smooth video
- Anything else is built sign by sign from the library

Patient → doctor

WORKING NOW

- Yes or No, answered by tapping a signed button
- Typed replies, spoken aloud on the doctor's device so his eyes stay on her
- Emergency alerts, a body drawing and a pain scale, all without words
- Every exchange recorded in a transcript on each device

The two devices exchange this directly. None of it passes through our server.

Accessibility and inclusion

Designed for a patient who may read nothing at all.

Nothing depends on reading

WORKING NOW

Questions are signed, answers are tapped. Even the Yes and No buttons carry their own sign clips.

The literacy question is asked, not assumed

WORKING NOW

It is shown in sign on the patient's own phone, so she answers it herself instead of a nurse guessing for her.

Nothing depends on colour alone

WORKING NOW

Each half of a shared screen carries a heading, a hint and a differently shaped mark, for bright sunlight and for colour blind staff.

Nothing depends on hearing

WORKING NOW

Vibration and visible state replace every sound cue. The app never relies on a chime.

The poorer patient is not downgraded

WORKING NOW

One phone or two, it is the same app and the same questions. The no-phone path is not a cut-down version.

Emergency is reachable first

WORKING NOW

The triage screen opens before any question is asked. You should not fill in a form to say you cannot breathe.

Technical fit and implementation

Built to survive a hospital network, not a conference Wi-Fi.

Peer to peer, by design

Once paired, the devices exchange the consultation directly. The server introduces them and then has nothing to do with it.

Offline after first use

The service worker keeps the clips on the device, so a sign plays in milliseconds with no network.

Honest failure

No TURN server, so some networks will not pair. The app says so and offers one shared screen rather than routing a consultation somewhere else.

Evidence

1,933

automated tests, both halves

61

written architecture decisions

108

reviewed clips, 13.2 MB served

Django and DRF on Render, React and Vite on Netlify, Postgres on Neon, clips on Cloudflare R2. One repository, two deployments.

Ghanaian language, and local fit

Ghanaian-language support

WORKING NOW

- English or Twi spoken in, Twi caption out, via the Khaya API
- Ghanaian Sign Language, filmed here with Deaf signers — not American Sign Language
- GhSL is one language nationwide, so the signs do not change from region to region
- Plural forms match their singular sign, because GhSL does not mark plurals
- Ga and Ewe extend through the same interface, but are not built

UX, usability and local fit

WORKING NOW

- Runs on the phone a ward already owns — no app store, no download, no new hardware
- 13.2 MB holds the whole library, so a weak network is a first-visit cost only
- A reload does not lose the visit; a refreshed device rejoins by itself
- Big targets, high contrast, and state that is always visible rather than inferred
- One screen or two, chosen by a nurse in one tap at the start of the visit

Adoption and continued use

The cheapest thing about it is the tenth hospital.

Nothing to install

A link, not a deployment. A ward starts using it in the time it takes to type an address.

Adding a sign is filming, not coding

A new sign is recorded, reviewed and imported. We added 44 this week without touching the app.

The library is shared

Every hospital draws on the same reviewed clips, so the second hospital costs almost nothing to serve.

It degrades honestly

Where it cannot help, it says so. Staff learn quickly what it is for, which is how a tool survives past week one.

Before a real ward

- Clinical sign-off on the library
- The rest of the alphabet filmed
- The 16 body-part signs filmed
- A TURN server for closed networks

PLANNED

Where you can use it, right now

Open it on your own phone while we are still standing here.

Doctor or nurse

tiemeghana.netlify.app

Patient

tiemeghana.netlify.app/join

No second phone?

Open the first link and choose “No, share this device”. The whole visit runs on one screen.

It is a web site, so it opens in Chrome, in Safari, in any browser, on a phone, a tablet or a laptop. Nothing to download and no app store. Open it once and tap “Add to Home Screen” and it behaves like an installed app, with the sign videos kept on the device.

The server never learns what was said.

A deaf patient can say where the pain is. A doctor can be sure she understood the medicine. And nobody had to find an interpreter.

It runs today, on the phones people already carry, in Ghanaian Sign Language filmed here. It is a pilot, it is not approved for clinical use yet, and we say so on every screen until it is.

Thank you. Questions?